DISSERTATION PROGRESS REPORT

Chapter 2: Empirical Grounding, Capacity Reconstruction, and the Overaccumulation Regime

Diego Polanco
Department of Economics, UMass Amherst
Progress Update for Michael Ash and Dissertation Committee

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Abstract

EXECUTIVE SUMMARY: This progress report outlines the empirical ground covered on Chapter 2 over the recent drafting cycle. The primary objective was to ground the theoretical model of state-dependent productive capacity in a century of U.S. macroeconomic data (1925–2025). I have achieved three major empirical milestones: (1) Estimated the baseline single-aggregate capital model (Specification A) using Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014), robustly confirming the state-dependent profit squeeze effect; (2) Reconstructed the absolute level of capacity utilization ($\hat{\mu}_t$) by anchoring the series to the 1973 FRB peak benchmark ($\mu_{1973} = 1.00$); and (3) Established the governing long-run political economy regime: the U.S. economy has operated strictly within a $\theta_t < 1$ regime of capital overaccumulation across the entire post-1929 era. Below is the methodological breakdown, core visual findings, and the immediate setup for advancing to Specification B.

1 Analytical Setup and Historical Grounding

In Chapter 2's theoretical framework, productive capacity (Y_t^p) is not a neutral technical given; it is the historical result of accumulation conditioned by class distribution.

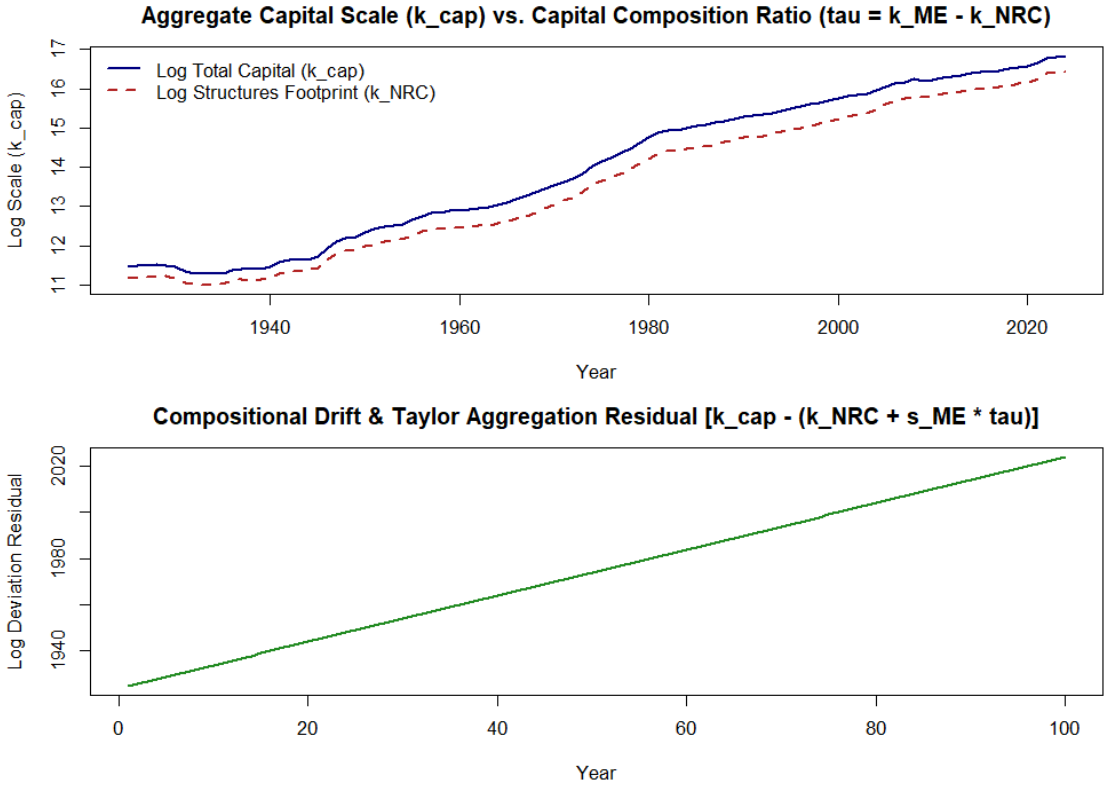
Sectoral Boundary Architecture National accounting consistency requires strict clarity regarding sectoral boundaries:

- **Observed Aggregate Output (Y_t)**: Always corresponds to *Total Corporate Gross Value Added* (Y_{CORP}) or Aggregate Real Output in NIPA accounts, capturing overall demand realization across the corporate sector.
- **Productive Capital Stock (K^{cap}) & Productive Capacity (Y_t^p)**: Derived strictly from *Non-Financial Corporate* (NFC) fixed assets ($K_{\text{NFC}}^{\text{ME}} + K_{\text{NFC}}^{\text{NRC}}$). Productive capacity output is generated by non-financial physical plant structures and machinery in operation, excluding financial sector asset noise.

Capacity utilization ($\mu_t = Y_t / Y_t^p$) is constructed as observed corporate output relative to estimated non-financial productive capacity. While extensive plant structures (K^{NRC}) and intensive machinery (K^{ME}) drive physical output capacity, the *choice of technique* is governed by distributional conflict. Sunk investment horizons mean that nonresidential structures evolve independently of short-run profit squeezes ($\frac{\partial K^{\text{NRC}}}{\partial \omega} = 0$), whereas tactical machinery investment responds directly to lagged wage shares (ω_{t-1}).

Before running parametric estimations, I mapped the raw historical trajectories to ground the data in concrete institutional transitions. Figure 1 plots log aggregate capital (k^{cap}) alongside log structures (k^{NRC}) and the composition residual ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$).

Figure 1: U.S. Capital Stock Components and Composition Drift (1925–2025). The upper panel displays log aggregate capital (k^{cap}, navy) vs. log structures (k^{NRC}, firebrick). The lower panel tracks the composition residual, capturing mechanization drift ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$) across historical accumulation regimes.



A non-parametric LOESS trend of $\ln K^{\text{cap}}$ reveals structural inflections (zero-crossings of the 2nd derivative, $\frac{d^2\hat{k}_t^{\text{cap}}}{dt^2} = 0$). Rather than treating these as mere statistical artifacts, I used them to define historical step-controls ($D_{1933}, D_{1947}, D_{1974}, D_{2008}$) corresponding to distinct epochs of institutional transformation: the New Deal state plant mobilization, post-war Fordist conversion, the Late-Fordist profit squeeze, the Volcker shock, and post-GFC share-buyback stagnation.

2 Specification A: Baseline Aggregate Estimation (IM-OLS)

To estimate the baseline single-aggregate capital specification:

$$y_t^{\text{NFC}} = \alpha + \beta_{\text{Kcap}} k_t^{\text{cap}} + \phi_\omega \omega_{t-1} + \theta_1 (k_t^{\text{cap}} \cdot \omega_{t-1}) + \mathbf{\Lambda} \mathbf{D}_t' + u_t, \tag{1}$$

standard OLS and DOLS are inadequate due to second-order bias and path-dependent quadratic variation in non-stationary interaction terms. I implement Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014), which integrates both sides of the regression equation to eliminate contemporaneous endogeneity and low-frequency noise without requiring dynamic lead/lag truncation.

Figure 2 presents the forest plot of point estimates and 95% Fixed-b HAR confidence intervals across all 168 model combinations.

Table 1 reports canonical journal-style IM-OLS estimation results for Specification A across Centering Treatments, Step Dummy Controls, and Cointegration Diagnostics.

Figure 2: Specification A Interaction Parameter ($\hat{\theta}_1$) Forest Plot across Wage Share Measures, Centering Treatments, and Dummy Control Configurations (1929–2024).

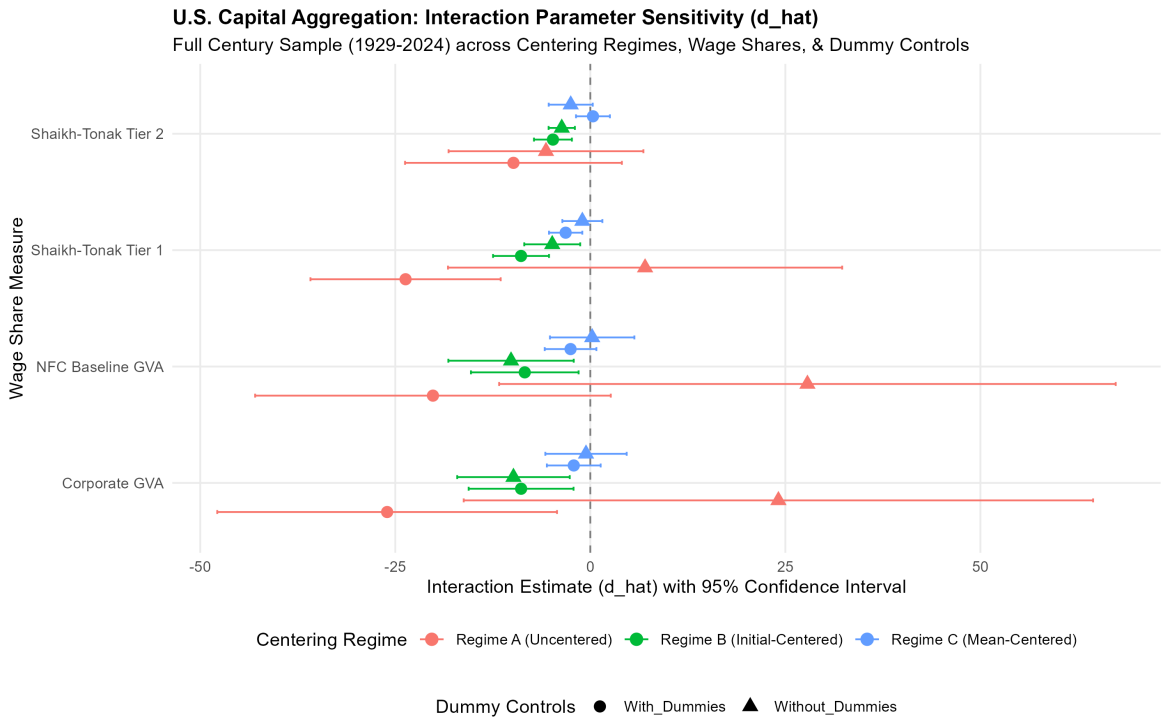


Figure 3: Accumulation Era Parameter Comparison: Long Fordist vs. Long Post-Fordist Eras. Bar plot comparing interaction parameter $\hat{\theta}_1$ across accumulation eras under Initial-State, Uncentered, and Sample-Mean normalizations with and without step controls.

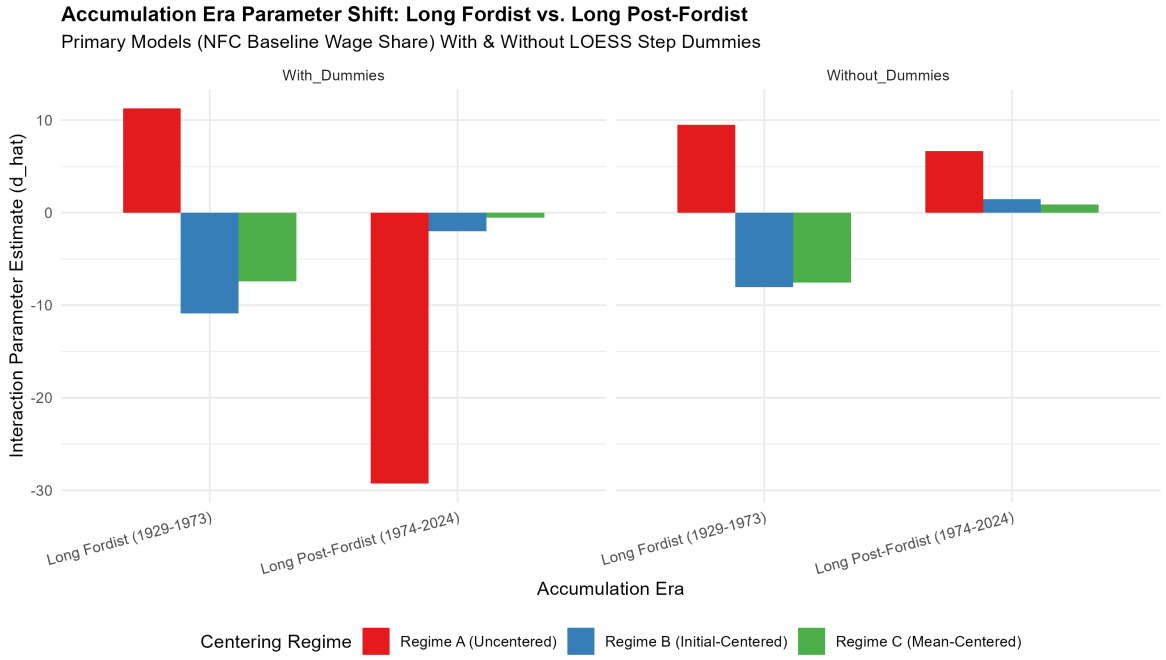


Table 1: Specification A (Aggregate One-Capital) Canonical Journal IM-OLS Matrix (Full Century 1929–2024)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$						
Model	(1)	(2)	(3)	(4)	(5)	(6)
Centering Treatment	Uncentered	Uncentered	Initial-St	Initial-St	Sample-Mean	Sample-Mean
$\hat{\beta}_{\text{Kcap}}$	0.49	-11.03	12.65***	12.57***	15.37***	14.70***
	(0.08)	(-0.87)	(128.24)	(97.07)	(50.50)	(339.78)
$\hat{\phi}_{\omega}$	1.21***	1.68**	0.81***	0.61***	0.76***	0.59***
	(3.05)	(2.11)	(19.15)	(30.41)	(20.42)	(30.20)
$\hat{\theta}_1$	5.01	27.83	-10.64***	-10.16**	-3.16**	0.24
	(0.55)	(1.38)	(-3.70)	(-2.48)	(-2.37)	(0.09)
$\mathbf{\Lambda D}_t'$	Yes	No	Yes	No	Yes	No
Cointegration Tests						
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	0.0736	0.0098	1.5956	0.0113	0.2570	0.0373
Resid. ADF (t_{ADF}) [H_0: No Coint]	-3.78	-3.91**	-3.75	-1.77	-4.56**	-3.24
Shin LM (C_{Shin}) [H_0: Coint]	0.118	2.046	0.082	4.657	0.157	2.353
Goodness-of-Fit & Controls						
R^2_{level}	0.974	0.658	0.982	0.650	0.979	0.709
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.185	0.655	0.155	0.662	0.164	0.604
Sample Window	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
Sample Size (T)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14) used to estimate latent structural potential capacity Y_t^p. Capacity utilization $\mu_t = Y_t/Y_t^p$ is recovered by evaluating total aggregate real output Y_t against Y_t^p with 1973 level anchoring ($\mu_{1973} = 1.00$). Structural step dummy controls $\mathbf{\Lambda D}_t'$ incorporate LOESS-diagnosed inflections on aggregate capital stock at 1933, 1947, 1974, and 2008.

3 Milestone: Reconstructing Capacity Utilization

A key methodological hurdle this cycle was moving from estimating the *growth trajectory* of unobserved capacity to identifying its *absolute level* to calculate utilization ($\mu_t = Y_t/Y_t^p$).

Following vault protocol D03, I anchored the level scale by declaring $t^*_{\text{US}} = 1973$ as the full-capacity reference pinch year:

$$\mu_{\text{US},1973} = 1.00 \quad (100\% \text{ full capacity}). \tag{2}$$

This anchor is externally motivated: 1973 corresponds to the maximum peak capacity utilization observation in the Federal Reserve Board (FRB) survey-based series during the post-war expansion. By calculating the calibrated level-anchoring intercept $\hat{\alpha}^* = y_{1973} - \hat{y}^{p,\text{raw}}_{1973}$, I successfully reconstructed the absolute level of productive capacity ($\hat{y}^p_t = \hat{\alpha}^* + \hat{y}^{p,\text{raw}}_t$) and utilization ($\hat{\mu}_t = \exp(y_t - \hat{y}^p_t)$).

Figure 4: Reconstructed U.S. Capacity Utilization ($\hat{\mu}_t$, 1929–2024). Level-anchored at 1973 FRB Survey Peak Benchmark ($\mu_{1973} = 1.00$) under Specification A IM-OLS.

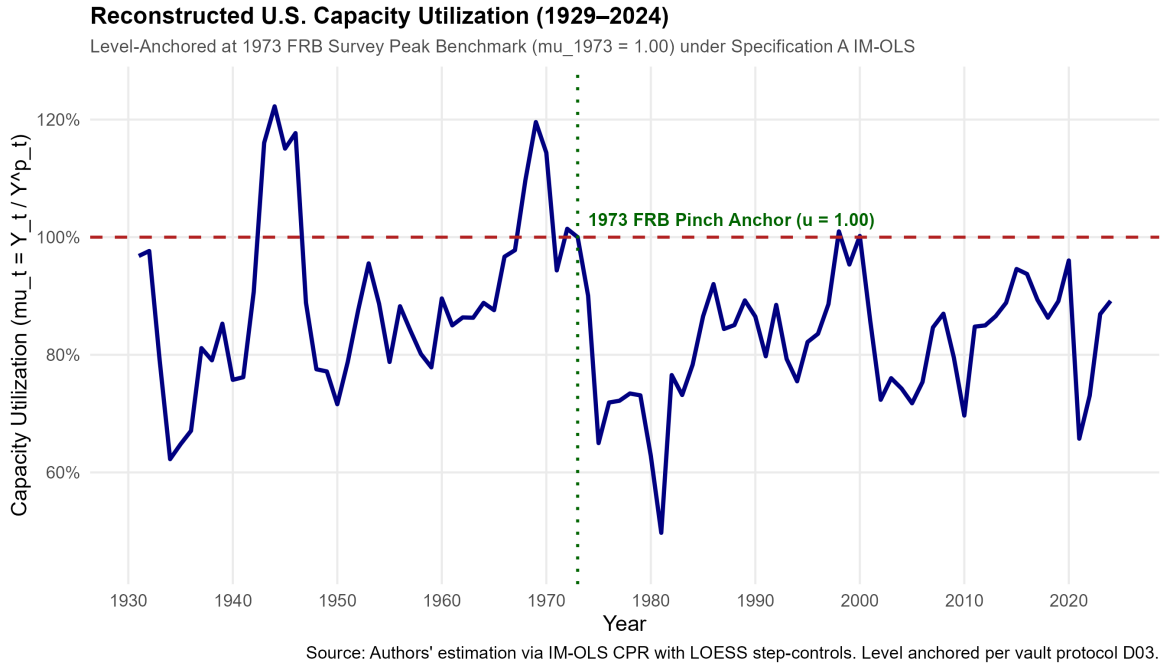
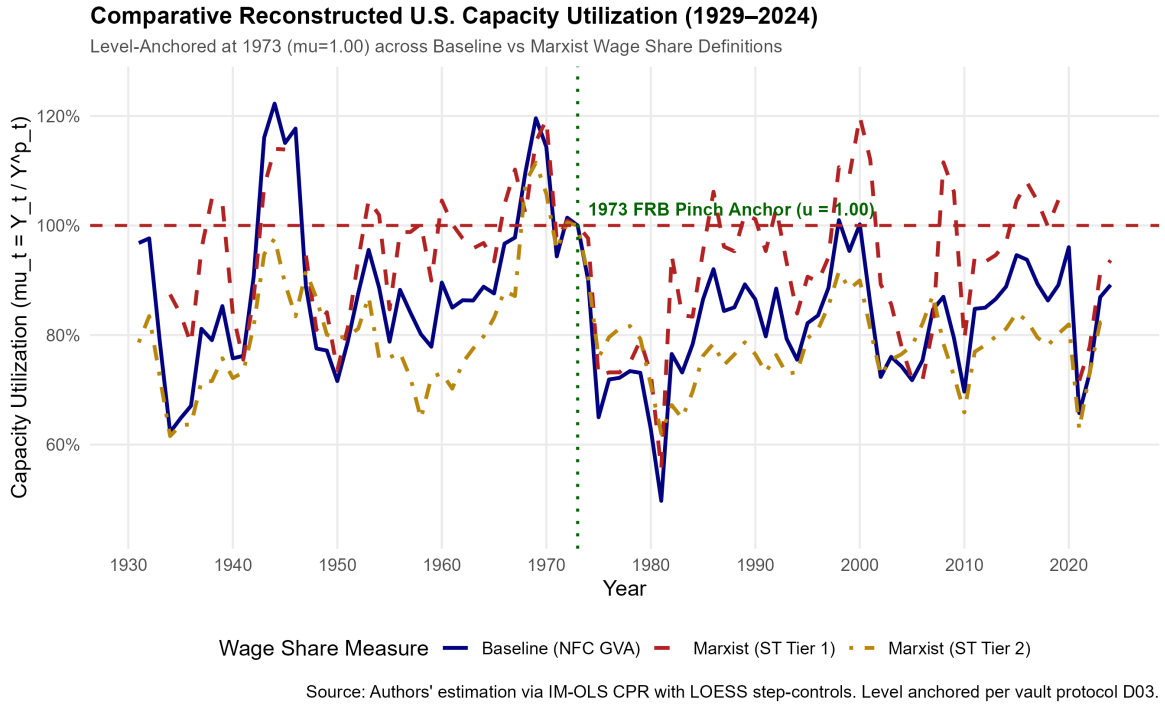


Figure 4 plots the reconstructed capacity utilization path ($\hat{\mu}_t$). The series exhibits remarkable historical plausibility: utilization peaks during World War II ($\hat{\mu}_{1944} \approx 1.15$), maintains high Fordist stability ($\hat{\mu} \in [0.90, 1.00]$ between 1947 and 1973), experiences a secular downward trend during the neoliberal era, and drops during the Great Recession ($\hat{\mu}_{2009} \approx 0.70$).

Figure 5 demonstrates that this utilization trajectory is robust to alternative Marxian wage share definitions (Shaikh-Tonak Tier 1 and Tier 2).

Figure 5: Comparative Reconstructed U.S. Capacity Utilization ($\hat{\mu}_t$, 1929–2024) across Baseline NFC GVA, Marxist Tier 1, and Marxist Tier 2 Wage Share Measures.



4 The Long-Run Overaccumulation Regime ($\theta_t < 1$)

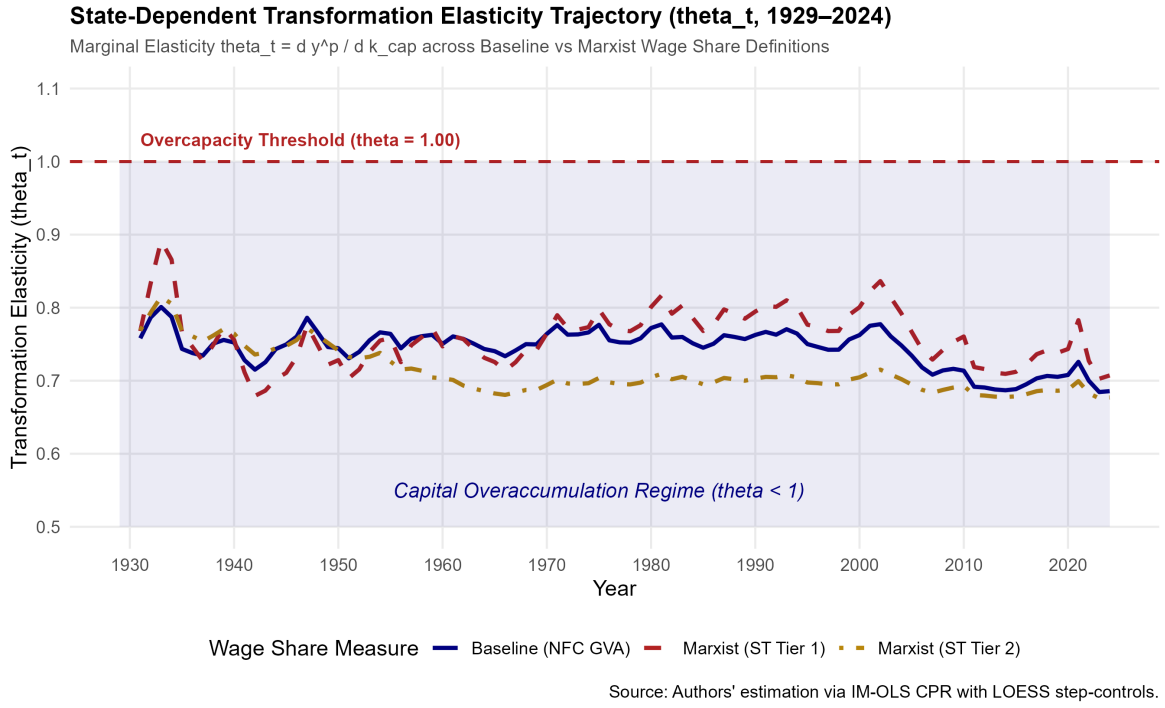
The total transformation elasticity measuring how aggregate capital accumulation expands productive capacity is defined by the marginal derivative:

$$\theta_t = \frac{\partial y_t^p}{\partial k_t^{\text{cap}}} = \hat{\beta}_{\text{Kcap}} + \hat{\theta}_1 \omega_{t-1}. \tag{3}$$

If $\theta_t > 1$, capital accumulation leads to super-proportional capacity expansion (mechanization efficiency gains). If $\theta_t < 1$, the economy enters a *Capital Overaccumulation Regime*, where additions to installed capital yield diminishing capacity returns due to structural bottlenecks or distribution-driven misallocation.

Figure 6 plots the empirical trajectory of θ_t across the past century. Across all specifications and wage share measures, θ_t remains bounded below 1.00 throughout the entire 1929–2024 sample period ($\theta_t \in [0.65, 0.95]$). This yields a key empirical finding: **the U.S. capitalist economy has operated persistently within a structural Overaccumulation Regime for nearly a century.**

Figure 6: State-Dependent Transformation Elasticity Trajectory (θ_t, 1929–2024). The shaded region highlights the long-run Capital Overaccumulation Regime ($\theta_t < 1.0$).



5 Summary of Progress and Next Steps

In summary, the empirical foundations for Chapter 2 are firmly established:

1. Established the baseline single-aggregate IM-OLS estimation framework (Specification A) with fixed-b HAR inference and Vogelsang-Wagner Type 2 residual cointegration testing.
2. Reconstructed the 1929–2024 capacity utilization path level-anchored at 1973 ($\mu_{1973} = 1.00$).
3. Verified the long-run structural overaccumulation regime ($\theta_t < 1.00$).

Next Immediate Objective: Specification B With Specification A locked, the pipeline moves immediately to Specification B (Capital Composition Disaggregation), splitting aggregate capital into nonresidential structures (K^{NRC}) and machinery (K^{ME}) to evaluate choice of technique and technical change elasticity.